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Dynamics and Vibrational Control of an Underwater Inverted Pendulum

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Abstract: This paper discusses the dynamics and vibrational control problem of an underwater inverted pendulum system. The system's equations of motion are obtained by using the Lagrange equation. The vibrational control technique is applied to stabilize the inverted pendulum in water. Furthermore, stability analyses in terms of stable-unstable region for the underwater inverted pendulum system is presented. While a number of control techniques for the stabilization problem of an inverted pendulum in air by oscillating the support point have been developed, but a similar technique for the system in water has not yet been investigated. The dynamics of any system in liquid is strongly influenced by the density of the liquid and the relative velocity of the surrounding liquid. Also, it becomes a challenging task to stabilize the system in the presence of the hydrodynamic forces acting on it. Simulation results reveal that the application of the vibrational control technique newly developed in this paper can successfully stabilize the system in water under the influence of hydrodynamic forces.

Keywords: Underwater inverted pendulum, vibrational control, oscillatory input, hydrodynamics forces.

1. INTRODUCTION

An inverted pendulum is a frequently-used dynamic model to design, analyze, and stabilize many existing mechanical systems like broom balancing, flight simulation of a rocket, and a missile during the initial flight stage. In such mechanical systems mentioned above, open loop control strategies by introducing oscillatory vibrations at the support point have been developed based on a model assuming that the motions occur in air. However, the design and control problem for similar underwater systems has not been addressed yet, in which case the oscillatory vibrations should be provided in underwater environment for the purpose of stabilization. The motion and control of the systems in water become challenging because of the effect of hydrodynamic forces.

In this paper, the dynamics and vibrational control problem of an underwater inverted pendulum system for the purpose of stabilization in water are investigated. For ensuring successful underwater operations, the stability of the system should be ensured under the influence of hydrodynamic forces. The Lagrange equation is used to model the dynamics of an inverted pendulum under the effect of hydrodynamic forces. Also, the vibrational control method in air (i.e., an open-loop technique) is extended for the stabilization of the underwater inverted pendulum system. The stability region for a given underwater inverted pendulum system is presented as well.

There are many scientific works, already existing in the literature, on the stabilization of statically unstable systems by parametric vertical excitation. It is already known that the upright unstable equilibrium point of an inverted pendulum can be stabilized by periodic forcing with sufficiently high frequency and low amplitude of the oscillatory input source [1]. In 1951, the Russian physicist Kapitza [2] presented an analysis on the stability of an inverted pendulum with a vertically oscillatory base, which was an iconic application of averaging theory applied on oscillatory stabilization. This is why the vertically driven inverted pendulum is now sometimes called the Kapitza pendulum. Initial work on developing a general theory of vibrational control was carried out by Meerkov [3]. He discussed the effect of vibrational control upon stability, transient motion, and response of the controlled system. It can be said that the vibrational control theory for linear finite dimensional systems has been established by him. Afterward, approaches to oscillatory stabilization based on averaging have received extensive attention across many disciplines including control theory.

Vibrational control is an open loop control technique, which is easy to implement than other control schemes. Many authors have introduced various systems and applied the vibrational control theory. After Meerkov's work on vibrational control, several authors have contributed to the development of open-loop oscillatory control theory and stability analysis on linear and nonlinear systems by using averaging [4, 9], without averaging [5], control of under-actuated mechanical systems [4, 6, 8], integro-differential equation [10], and infinite dimensional systems [11-15]. In general, vibration control methods require known plant structures compared to other control methods including adaptive control and boundary control [7,15-25].

Particularly, in [8], vibrational control for under-actuated mechanical systems has been applied on a two link robotic arm. In the said work, the author have presented an open-loop oscillatory control with low amplitude and high frequency modulation. By first positioning the actuated robotic arm joints in their desired positions, vibrational input was applied to the actuated joints to bring the un-actuated joints to their desired locations. In [5], the authors have shown that the

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vibrational control theory can also be applied on the mechanical systems without using the method of averaging. They showed that the vibrational control theory can also be applied via stability maps. In [6], the author has discussed that the motion of an autonomous underwater vehicle (AUV) can be controlled with reduced control schemes in the case of an actuator failure. He described a technique for synthesizing controls for under-actuated AUVs and showed how to use his technique to provide adaptation to changes in control authority. The vibrational control theory was extended to parabolic partial differential equations in [10]. Vibrational control of nonlinear time lag systems was discussed in [11]. A geometrical picture is given in [26] that explains the phenomenon of stabilizing an inverted pendulum by applying high frequency oscillations. A pendulum based model was used to represent fluid-structure analysis in [27] and a theoretical and experimental analysis of the dynamic behavior of a viscously damped spherical non-rotational one-degree-of-freedom pendulum was performed.

In [28], the authors have presented another scheme for the vibrational control of under-actuated mechanical systems based on the use of geometric control and concepts in differential geometry. The residual vibration control problem of a nuclear power plant's fuel transport system (i.e., an underwater system) was presented in [29]. In [30], the authors presented a control strategy for harvesting energy from a vertically excited parametric underwater pendulum.

In this paper, the dynamics of an underwater inverted pendulum system subject to an oscillatory input along the vertical direction for stabilization purpose is studied. The mathematical model of the system is derived in consideration of the incident hydrodynamic forces and the oscillatory input at the pivot point of the inverted pendulum. The Lagrangian mechanics has been used to derive the equations of motion of the underwater system. Furthermore, the stability analysis in the form of stability region is given for an underwater inverted pendulum system.

The rest of the paper is organized as follows. The physical system and problem formulation are introduced in Section 2. The mathematical model of the underwater inverted pendulum system is formulated in Section 3. Vibrational control of the underwater inverted pendulum system is presented in Section 4. Simulations of the uncontrolled and controlled systems are presented in Section 5. Finally, the conclusions are made in Section 6.

2. PHYSICAL SYSTEM AND PROBLEM FORMULATION

Fig. 1 shows the considered underwater inverted pendulum system in this paper. The pendulum system has two equilibrium points: One is the upward unstable equilibrium position (i.e., 0°) and the other is downward stable equilibrium position (i.e., 180°). It is emphasized again that the pendulum operates in the underwater environment. Oscillatory vibrations have been introduced vertically at its pivot point to stabilize the system and keeping it stable at its upward unstable equilibrium position. The pendulum is directly attached to the support point, which produces oscillations according to our desired input amplitude and forcing frequency. Initially (i.e., at 0 s), the pendulum is displaced on its vertical unstable equilibrium position to study the effect of hydrodynamics forces acting on its outer surface. An input vibrational source is producing the oscillations with desired smaller amplitude and sufficiently higher frequency to keep it at its upward unstable equilibrium position.

Since the inverted pendulum system considered here is in an underwater environment, the hydrodynamic forces (i.e., the buoyancy force F_B and the drag force F_{Drag}) acting on its surface will affect the dynamic behavior of the inverted pendulum. The drag force F_{Drag}, in-line with the flow, can be calculated as follows.

$$F_{Drag} = \frac{1}{2} C_D \rho_w A_p v^2, \tag{1}$$

where C_D is the drag force coefficient, A_p is the pendulum's projected area, ρ_w is the fluid density, and v is the relative velocity between the fluid and inverted pendulum. In this paper, the water is static and only the velocity v of the inverted pendulum is considered.

The force due to buoyancy can be described mathematically as

$$F_B = \rho_w V_r g, \tag{2}$$

where V_r is the volume of the pendulum and g is the gravitational acceleration force.

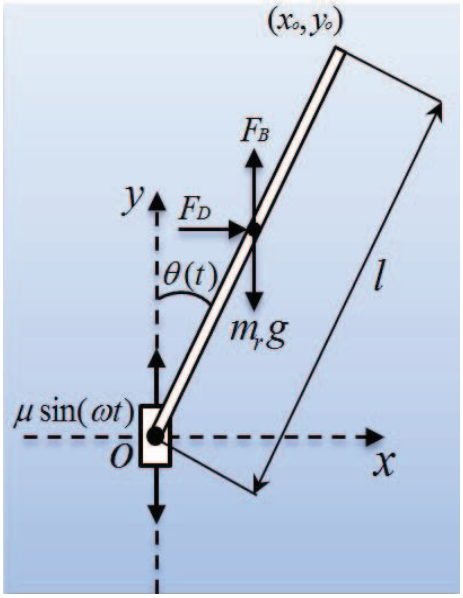

Fig. 1 Underwater inverted pendulum system with vibrational input source.

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When a body moves in the fluid, it carries its own mass and the added mass of the fluid. This added mass provides an additional (or extra) inertia in the system. So the added mass is given as follows.

$$m_a = C_a \rho_w V_r, \tag{3}$$

where C_a is the added mass coefficient.

3. MATHEMATICAL MODEL

The dynamical model of the underwater inverted pendulum system is derived using the Lagrange equation. Inverted pendulum is free to move around its point of suspension. Thus the system has one degree of freedom. $\theta(t)$ is considered to be the generalized coordinate. The dynamic equation of an underwater inverted pendulum system from Fig. 1 can be derived as follows.

$$x_o = l \sin\theta, \tag{4}$$

$$y_o = l \cos\theta + \mu \sin \omega t. \tag{5}$$

The kinetic energy (T) and the potential energy (U) of the underwater inverted pendulum system are given as follows.

$$T = \frac{1}{2} m v^2, \tag{6}$$

$$U = (m_r g - F_B) y, \tag{7}$$

where v is the velocity of the pendulum. The velocity is a vector representing the change in the position given as

$$v^2 = l^2 \dot{\theta}^2 - 2\mu l \omega \sin\theta \cos(\omega t)\dot{\theta} + \mu^2 \omega^2 \cos^2(\omega t). \tag{8}$$

The viscous damping coefficient associated with the movement of the inverted pendulum system in water can be represented in form of dissipation energy (D) as follows.

$$D = \frac{1}{2} D_\theta \dot{\theta}^2, \tag{9}$$

where D_θ is the viscous damping coefficient associated with the motion of the underwater inverted pendulum. The drag force is considered to be the only generalized force acting on the system, which is given as follows.

$$Q = \pm F_D. \tag{10}$$

The signal of the drag force changes according to the angular movement of the inverted pendulum [16]. If the angular motion is in clockwise direction then the drag force is positive, otherwise it is negative.

Substitution of Eqs. (6)-(10) into the Lagrange equation yields the following equation of motion of the underwater inverted pendulum system, including the response of the oscillatory input source.

$$\ddot{\theta} = \frac{1}{ml^2}(((m_r g - F_B)\, l - m\mu l \omega^2 \sin(\omega t)) \sin\theta - D_\theta \dot{\theta} \pm F_D), \tag{11}$$

where $m = m_r + m_a$, m_r is the pendulum mass, l is the length of the pendulum, μ is the excitation amplitude and ω is the forcing frequency of the vibrational input source. The above nonlinear ordinary differential equation of second degree fully describes the dynamic behavior of the underwater inverted pendulum system. The dynamic motion and vibrational control is presented in the next section.

4. VIBRATIONAL CONTROL

This section describes the vibrational control technique of an underwater inverted pendulum for the purpose of stabilization at the unstable upward equilibrium position. The pendulum exhibits the dynamics of a second–order nonlinear system. In vibrational control, oscillations of small excitation amplitude along with high forcing frequency are introduced at the pivot point. The excitation amplitude of the induced oscillations should be very small as compared with the length of the pendulum. On the other hand, the frequency should be significantly large as compared with the natural frequency of the system.

From Eq. (11), assuming $\sin\theta \approx \theta$, the linearized equation is obtained as follows.

$$\ddot{\theta} = \frac{1}{ml^2}\begin{pmatrix} \left((m_r g - F_B) l - m\mu l \omega^2 \sin(\omega t)\right)\theta \\ - D_\theta \dot{\theta} \pm F_D \end{pmatrix}. \tag{12}$$

By applying the averaging principle in [3], the equation with constant coefficient obtained from Eq. (12) is obtained as follows.

$$\ddot{y} = \left(\frac{(m_r g - F_B)}{ml} - \frac{\mu^2 \omega^2}{2l^2}\right) y - \frac{D_\theta \dot{y}}{ml^2} \pm \frac{F_D}{ml^2}. \tag{13}$$

Equation (13) shows that if constant oscillations of very low excitation amplitude and high forcing frequency are introduced according to the law $\mu\omega^2 \sin(\omega t)$ with the condition

$$\mu^2 \omega^2 > 2(m_r g - F_B) l / m, \tag{14}$$

then the induced vertical vibrations at the support point asymptotically stabilize the averaged system given by Eq. (13).

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5. SIMULATIONS

This section illustrates the vibrational control and stability analysis of an underwater inverted pendulum system via numerical simulations. A fourth-order Runge Kutta method is employed to compute the numerical solutions of Eq. (11). To study the dynamic response and vibrational control of an underwater inverted pendulum system, the numerical simulations are carried out by using MATLAB's Simulink Toolbox. In the first phase, the fluid (water) is at rest. At 0 s, the inverted pendulum is displaced by 0.01 rad, relative to the vertical axis, from its unstable upward equilibrium position without applying any vibrational input in water. The time response of the angular displacement of the underwater system is shown in Fig. 2. It is observed that the angular displacement increases and pendulum moves away from its unstable equilibrium position as time increases and finally becomes stable at the stable downwards equilibrium position at 180°. It also verifies that the mathematical model derived in Eq. (11) is correct. Fig. 3 shows the undamped behavior of a simple inverted pendulum system in air, which shows the ideal dynamic response. The angular velocity profile without any vibrational input for the underwater pendulum system is presented in Fig. 4.

In the second phase, an underwater pendulum system is displaced from its unstable upward equilibrium position by 0.01 rad. Now the pivot point of the pendulum system is given constant vibrations in the vertical direction for the purpose of stabilization. A vibrational input of a very small excitation amplitude (i.e., μ = 0.1 m) along with a high forcing frequency of ω = 70 rad is applied at the pivot point. It is observed that the pendulum system has quickly become stable around its unstable equilibrium position at the upward position at 0° regardless of the opposing effect of the hydrodynamic force acting on the system. This response is shown in the Fig. 5. It shows that periodic vibrations induced at the pivot point stabilized the system from where it was initially displaced. It also shows that the induced periodic vibrations controlled the movement of the underwater inverted pendulum and keeps it within a certain boundary until the movements become zero.

Table 1 Simulation parameters

Parameters	Units	Value
Mass of the pendulum (m_r)	Kg	0.165
Length of the pendulum (l)	m	1
Diameter of the pendulum (d)	m	0.01
Drag coefficient (C_D)	-	1.28
Added mass coefficient (C_a)	-	2
Water density (ρ_w)	Kg/m^3	1000
Viscous damping coef. (D_θ)	Nm-s/ rad	0.04

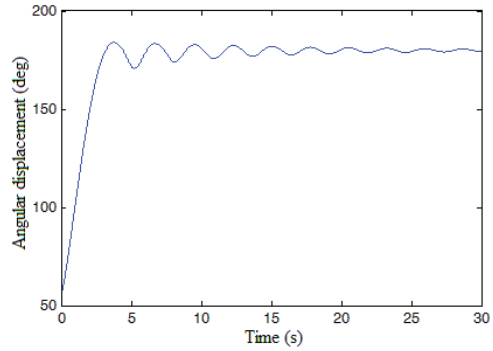

Fig. 2 Angular displacement of the pendulum in water with $\theta(0)$ = 1 rad (no control case).

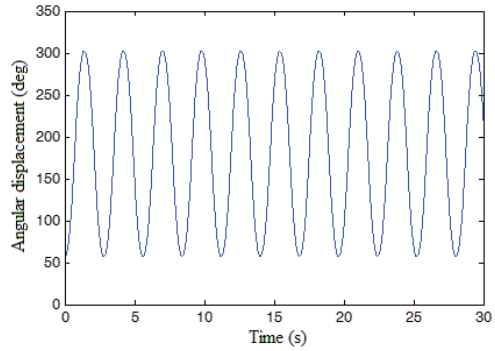

Fig. 3 Angular displacement of the pendulum in air with $\theta(0)$ = 1 rad (no control case).

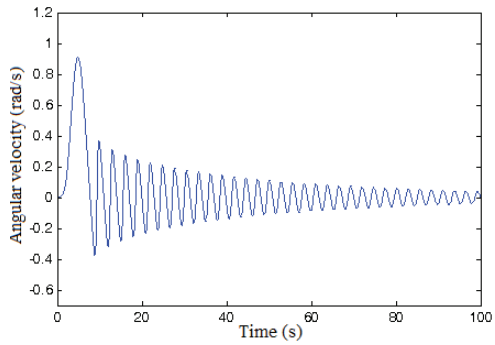

Fig. 4 Angular velocity of the pendulum in water in the case of Fig. 1 (no control).

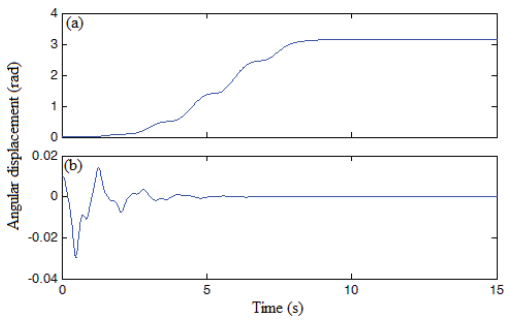

Fig. 5 Angular displacement of the pendulum in water under the vibration control when $\theta(0)$ = 0.01 rad: (a) ω = 4, μ = 0.4; (b) ω = 16, μ = 0.4.

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Fig. 6 presents the stability region of the inverted pendulum in water. It shows that the underwater inverted pendulum system can be stabilized for bounded values of excitation amplitudes and forcing frequencies within the region. It further illustrates that the system is more stable above the solid line. Any combination of values of excitation amplitude and forcing frequency that lies above the solid line tends to stabilize the system. Fig. 7 compares the angular displacements of the underwater pendulum with various magnitude and frequency conditions.

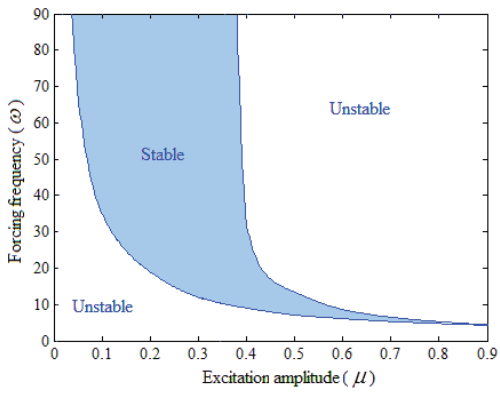

Fig. 6 Stability region of the underwater inverted pendulum system.

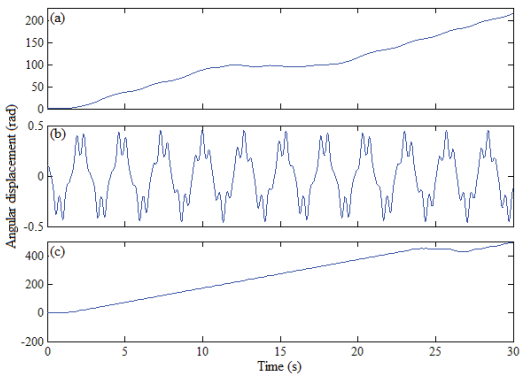

Fig. 7 Angular displacements of the underwater inverted pendulum: (a) $\omega = 10$, $\mu = 0.2$; (b) $\omega = 14$, $\mu = 0.4$; (c) $\omega = 20$, $\mu = 0.6$.

6. CONCLUSIONS

In this paper, the dynamic response and stabilization through vibrational control of an unstable underwater inverted pendulum was presented. Results showed that when periodic vibrations are introduced at the pivot point of an underwater inverted pendulum system, the response tends to become stable around its upright unstable equilibrium position (i.e., 0°). Furthermore, the stability region for the underwater system was also presented. This illustrates that any combination of excitation amplitude and forcing frequency that lies with the stability region induces the stabilization of the system in water.

ACKNOWLEDGMENT

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